GradeWise Service

Automated Exam Evaluation using AI

Question Paper › Model Answer › Student Sheet › 4 Results

Start New Evaluation

Evaluation Result

Student ID: MH002

F

15.0 / 40.0

Scoreboard

Question	Score	Max	%	Status
Section A				
Q1	1	1	100% 1/1	Scored
Q2	1	1	100% 1/1	Scored
Q3	1	1	100% 1/1	Scored
Q4	1	1	100% 1/1	Scored
Q5	1	1	100% 1/1	Scored
Q6	0	1	0% 0/1	Not Scored
Q7	1	1	100% 1/1	Scored

Question	Score	Max	%	Status
Q8	1	1	100% 1/1	Scored
Section B				
Q9	1	2	50% 1/2	Scored
Q10	0	2	0% 0/2	Not Scored
Q11	1	2	50% 1/2	Scored
Q12	2	2	100% 2/2	Scored
Q13	1	2	50% 1/2	Scored
Q14i	0	1	0% 0/1	Not Scored
Q14ii	0	1	0% 0/1	Not Scored
Section C				
Q15	0	3	0% 0/3	Not Scored
Q16i	0	1.5	0% 0/1.5	Not Scored
Q16ii	0	1.5	0% 0/1.5	Not Scored
Q17	1.5	3	50% 1.5/3	Scored
Q18	1.5	3	50% 1.5/3	Scored
Section D				

Question	Score	Max	%		Status	
Q19	0	8	0%	0/8	Not Scored	
Total	**15**	**40**	**38%**		—	—

Question-wise Details

Question 1

1 / 1

The student's answer matches the model answer exactly in terms of the correct option and value. Both indicate that the number of subsets of A is 8, which is the correct answer for a set with 3 elements (2^3 = 8).

Suggestions

- Ensure to include the option letter and value as shown in the model answer for clarity.
- Maintain consistent formatting with the question's options to avoid any ambiguity.

Question 2

1 / 1

The student's answer matches the model answer exactly in meaning and content, correctly identifying the universal set as the set containing all elements under consideration. Minor differences in formatting (such as the absence of parentheses) do not affect correctness.

Suggestions

- Ensure to match the formatting of the options as given in the question (e.g., include parentheses around the option letter) for full presentation accuracy.

Question 3

1 / 1

The student's answer matches the model answer exactly in terms of the correct option and value. Both indicate that cos A is 4/5, which is the correct value given sin A = 3/5 for an acute angle A.

Suggestions

- Ensure to include the reasoning or calculation steps to demonstrate understanding, such as using the Pythagorean identity: cos A = sqrt(1 - sin^2 A) = sqrt(1 - (3/5)^2) = 4/5.
- Write the answer in the same format as the question options for clarity, e.g., '(a) 4/5'.

Question 4

1 / 1

The student's answer 'c) Many-one relation' matches the model answer '(c) Many–one relation' exactly in meaning and choice, despite minor formatting differences. Therefore, full marks are awarded.

Suggestions

- Ensure consistent use of formatting and symbols, such as using en dashes (–) instead of hyphens (-), for better presentation.
- Include the option letter in the same format as the question (e.g., '(c)') to maintain uniformity.

Question 5

1 / 1

The student's answer matches the model answer exactly in terms of the option selected and the value of sin 30°, which is 1/2. The slight difference in formatting (parentheses around the option letter) does not affect correctness.

Suggestions

- Maintain consistent formatting with the question options, such as including parentheses around the option letter, for clarity.

Question 6

0 / 1

The correct number of ways to arrange 3 books out of 5 different books is 60, which corresponds to option (c). The student selected option (a) 10, which is incorrect. Therefore, no marks can be awarded.

Suggestions

- Review the concept of permutations, specifically nPr, which calculates the number of ways to arrange r items out of n distinct items.
- Recall that the formula for permutations is nPr = n! / (n - r)!; for 5 books arranged 3 at a time, it is 5P3 = 5 × 4 × 3 = 60.
- Carefully read all options before selecting the answer to avoid choosing an incorrect option.

Question 7

1 / 1

The student's answer matches the model answer exactly in meaning. The student correctly identified option (b) and used the symbol '∅' to represent the empty set, which is synonymous with '∅'. Therefore, the answer is fully correct.

Suggestions

- Use the standard empty set symbol '∅' to maintain consistency with common mathematical notation.
- Include the option letter in parentheses as in the model answer for clarity, e.g., '(b)'.

Question 8

1 / 1

The student's answer 'c) Transitive' matches the model answer '(c) Transitive' exactly in meaning and correctness. The relation 'less than' on natural numbers is indeed transitive, and the student correctly identified this property.

Suggestions

- Ensure to include the parentheses around the option letter to match the model answer format, e.g., '(c) Transitive'.
- Maintain consistent formatting for clarity, especially in formal assessments.

Question 9

1 / 2

The student's answer correctly identifies a set as a collection of objects, which captures the essence of the definition. However, it lacks the important qualifier 'well-defined' which is crucial in the formal definition of a set. Additionally, the student did not provide an example as requested. Therefore, the answer is partially correct but incomplete.

Suggestions

- Include the term 'well-defined' to emphasize that the collection must be clearly specified.
- Provide an example of a set, such as A = {1, 2, 3}, to fully answer the question.
- Use curly brackets '{}' instead of generic 'brackets' to denote a set.

Question 10

0 / 2

The student's answer correctly identifies the domain as all real numbers, which matches the model answer. However, the student incorrectly states that the range is also all real numbers, whereas the model answer specifies the range as all real numbers greater than or equal to zero ($x \geq 0$). Since the range is incorrect and no steps or reasoning were provided, no partial credit can be awarded.

Suggestions

- Clearly specify the range of the function $f(x) = x^2$ as all real numbers greater than or equal to zero.
- Use correct notation and terminology, for example, 'Domain: $\mathbb{R}$' and 'Range: $\{x \in \mathbb{R} \mid x \geq 0\}$'.
- Show understanding that squaring any real number results in a non-negative value.

Question 11

1 / 2

The student provided the correct final numerical answer of 1/2 but did not show any intermediate steps or working. According to the grading rules, this warrants half marks.

Suggestions

- Include the intermediate steps showing the values of sin 45° and cos 45° (both equal to $1/\sqrt{2}$) and their multiplication to reach the final answer.
- Use consistent notation (e.g., lowercase 'sin' and 'cos') for clarity.

Question 12

2 / 2

The student correctly calculated the value of $^5P_4$ as 5 × 4 = 20, which matches the model answer's numerical value. Although the notation 'SP2' is incorrect and does not match the standard permutation notation, the calculation and final answer are accurate and complete.

Suggestions

- Use the correct permutation notation, such as $^5P_4$, to clearly indicate the problem being solved.
- Ensure clarity in notation to avoid confusion, especially in formal mathematical contexts.

Question 13

1 / 2

The student provided the correct final answer (16) but did not show the steps or reasoning behind it. According to grading rules, correct final answer without working merits half marks.

Suggestions

- Include the formula for the number of elements in the power set, which is 2^n where n is the number of elements in the original set.
- Show the calculation explicitly: since A has 4 elements, the power set has 2^4 = 16 elements.

Question 14i

0 / 1

The student's answer does not provide a single example of a one-one function as requested. Instead, it mentions two functions and classifies them as one-one and many-one, but the example for one-one function given is f(n) = n, which is acceptable but not exactly matching the model answer 'f(x) = x + 1'. However, the question requires one example of a one-one function, and the student provides one. Despite that, the answer includes extra information about many-one functions, which is not asked for. Since the student's example f(n) = n is a valid one-one function, but the model answer is 'f(x) = x + 1', and the instructions require exact or synonymous answers for full marks, the answer is considered different. Given the instructions, the student did provide a correct example of a one-one function, but the answer is not exactly matching the model answer. However, the instructions do not explicitly say that alternative correct examples are unacceptable. Since $f(n) = n$ is a one-one function, it is an acceptable synonymous example. Therefore, the student answer is correct but includes unnecessary information. According to the instructions, the answer must match exactly or be synonymous to get full marks. Since the student provided a synonymous example ($f(n) = n$) for one-one function, the answer is correct but no steps are shown. Hence, half marks should be awarded.

Suggestions

- Provide only the required example without extra unrelated information.
- Use notation consistent with the question and model answer (e.g., use $f(x)$ instead of $f(n)$).
- Focus on answering precisely what is asked.

Question 14ii

0 / 1

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 15

0 / 3

The student's answer only rewrites the left-hand side expression without performing any algebraic manipulation or steps toward proving the identity. There is no simplification, no use of trigonometric identities, and no demonstration that the left-hand side equals the right-hand side. Therefore, the answer is incomplete and incorrect with no evidence of understanding or progress toward the proof.

Suggestions

- Show step-by-step algebraic manipulation to combine the terms on the left-hand side.
- Use common denominators to add the fractions properly.
- Apply Pythagorean identities such as $\sin^2\theta + \cos^2\theta = 1$ to simplify expressions.
- Demonstrate how the expression simplifies to $2 \operatorname{cosec}\theta$ explicitly.
- Write a concluding statement that the left-hand side equals the right-hand side, hence proving the identity.

Question 16i

0 / 1.5

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 16ii

0 / 1.5

No student answer available.

Suggestions

- Provide an answer to receive credit.

Question 17

1.5 / 3

The student provided the correct final answer (60) and correctly identified the permutation notation (5P3). However, the student did not show the calculation steps (5 × 4 × 3) as in the model answer. According to the grading rules, this warrants half marks.

Suggestions

- Include the calculation steps to show how the final answer is derived (e.g., 5 × 4 × 3).
- Explain briefly why permutation is used (digits without repetition).

Question 18

1.5 / 3

The student correctly identified that the relation is reflexive and symmetric, which matches part of the model answer. However, the student omitted the transitive property, which is also true for this relation and was included in the model answer. Since the answer is incomplete but partially correct, half of the total marks are awarded.

Suggestions

- Include all relevant properties of the relation: reflexive, symmetric, and transitive.
- Explain briefly why each property holds to demonstrate understanding.
- Mention that the relation is an equivalence relation if all three properties are satisfied.

Question 19

0 / 8

The student's answer only restates the left-hand side of the identity without any steps, reasoning, or proof. There is no demonstration of the equality or any algebraic manipulation to show the identity holds. Therefore, no credit can be awarded.

Suggestions

- Show the algebraic steps to transform the left-hand side into the right-hand side.
- Use the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$ to simplify the expression.
- Explain each step clearly to demonstrate understanding of the trigonometric identities involved.